Section 5.1 Matrices and Linear Systems

Matrices are mathematical objects and we can do fun things like add them and multiply them. The set of $n \times n$ matricies itself makes up something called a non-communitave Ring, not a topic for this class but worth a google if you are intrested. A Ring, in short, is an algebraic structure for which it's members satisify specific properties.

So we know how to add vectors but how we add matrices? If $A, B \in R^{n \times m}$, the set of $n \times m$ matrices.

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix} \qquad B = \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ b_{21} & b_{22} & \cdots & b_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ b_{n1} & b_{n2} & \cdots & b_{nn} \end{bmatrix}$$

Then

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \cdots & a_{1m} + b_{1m} \\ a_{21} + b_{21} & a_{22} + b_{22} & \cdots & a_{2m} + b_{1m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} + b_{n1} & a_{n2} + b_{n2} & \cdots & a_{nn} + b_{nm} \end{bmatrix}$$

That is we add them component wise.

Example: Calculate $A + B = \begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 2 \\ 2 & 7 & 4 \\ 0 & 8 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 4 & 3 \\ 6 & 1 & 0 \\ 2 & 5 & 2 \\ 9 & 1 & 10 \end{bmatrix}$

What about $B + A$?

Section 5.1 Matrices and Linear Systems

Scalar Multiplication

Let $c \in R$ and $A \in R^{n\times m}$ be a scalar then

$$cA = cA = c\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1m} \\ a_{21} & a_{22} & \cdots & a_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix} = \begin{bmatrix} ca_{11} & ca_{12} & \cdots & ca_{1m} \\ ca_{21} & ca_{22} & \cdots & ca_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ ca_{n1} & ca_{n2} & \cdots & ca_{nm} \end{bmatrix}$$

Example: Calculate

$$4\begin{bmatrix} 1 & 2 & 3 \\ -1 & 0 & 2 \\ 2 & 7 & 4 \\ 0 & 8 & 1 \end{bmatrix}$$

Theorem Let s and t be scalars and $A, B, C \in R^{n\times m}$ and 0_{nm} a matrix of zeros of size $n \times m$ then,

$(a)\ A + B = B + A$
$(b)\ s(A + B) = sA + sB$
$(c)\ (s + t)A = sA + tA$
$(d)\ (A + B) + C = A + (B + C)$
$(e)\ (st)A = s(tA)$
$(f)\ (A + 0_{nm}) = A$

Section 5.1 Matrices and Linear Systems

Matrix Multiplication

Let $A = [\vec{a}_1 \;\; \vec{a}_2 \;\; \cdots \;\; \vec{a}_k] \in R^{n\times k}$ whre $\vec{a}_i \in R^n$ and $B = [\,\vec{b}_1 \;\; \vec{b}_2 \;\; \cdots \;\; \vec{b}_m] \in R^{k\times m}$ and $\vec{b}_i \in R^k$ then the matrix product is defined by:

$$AB = \left[A\vec{b}_1 \;\; A\vec{b}_2 \;\; \cdots \;\; A\vec{b}_m\right]$$

$$= \left[(b_{11}\vec{a}_1 + b_{21}\vec{a}_2 + \cdots + b_{k1}\vec{a}_k) \;+\; (b_{12}\vec{a}_1 + b_{22}\vec{a}_2 + \cdots + b_{k2}\vec{a}_k) + \cdots + (b_{1m}\vec{a}_1 + b_{2m}\vec{a}_2 + \cdots + b_{km}\vec{a}_k)\right]$$

$$= \left[\left(b_{11}\begin{bmatrix}a_{11}\\a_{21}\\\vdots\\a_{n1}\end{bmatrix} + b_{21}\begin{bmatrix}a_{12}\\a_{22}\\\vdots\\a_{n2}\end{bmatrix} + \cdots + b_{k1}\begin{bmatrix}a_{1k}\\a_{2k}\\\vdots\\a_{nk}\end{bmatrix}\right) + \left(b_{12}\begin{bmatrix}a_{11}\\a_{21}\\\vdots\\a_{n1}\end{bmatrix} + b_{22}\begin{bmatrix}a_{12}\\a_{22}\\\vdots\\a_{n2}\end{bmatrix} + \cdots + b_{k2}\begin{bmatrix}a_{1k}\\a_{2k}\\\vdots\\a_{nk}\end{bmatrix}\right) + \cdots + \left(b_{1m}\begin{bmatrix}a_{11}\\a_{21}\\\vdots\\a_{n1}\end{bmatrix} + b_{2m}\begin{bmatrix}a_{12}\\a_{22}\\\vdots\\a_{n2}\end{bmatrix} + \cdots + b_{km}\begin{bmatrix}a_{1k}\\a_{2k}\\\vdots\\a_{nk}\end{bmatrix}\right)\right]$$

$$= \left[\begin{bmatrix}b_{11}a_{11} + b_{21}a_{12} + \cdots + b_{k1}a_{1k}\\b_{11}a_{21} + b_{21}a_{22} + \cdots + b_{k1}a_{2k}\\\vdots\\b_{11}a_{n1} + b_{21}a_{n2} + \cdots + b_{k1}a_{nk}\end{bmatrix} \;\; \begin{bmatrix}b_{12}a_{11} + b_{22}a_{12} + \cdots + b_{k2}a_{1k}\\b_{12}a_{21} + b_{22}a_{22} + \cdots + b_{k2}a_{2k}\\\vdots\\b_{12}a_{n1} + b_{22}a_{n2} + \cdots + b_{k2}a_{nk}\end{bmatrix} \;\; \cdots \;\; \begin{bmatrix}b_{1m}a_{11} + b_{2m}a_{12} + \cdots + b_{km}a_{1k}\\b_{1m}a_{21} + b_{2m}a_{22} + \cdots + b_{km}a_{2k}\\\vdots\\b_{1m}a_{n1} + b_{2m}a_{n2} + \cdots + b_{km}a_{nk}\end{bmatrix}\right]$$

We can also state this in the following way if $AB = C$ then

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + a_{i3}b_{3j} + \cdots + a_{ik}b_{kj}$$

Which is an $n \times m$ matrix

Example: Calculate $AB = \begin{bmatrix}1 & 0\\-1 & 2\\3 & 1\end{bmatrix}\begin{bmatrix}2 & -2 & 1 & 1\\4 & 2 & -1 & 0\end{bmatrix}$

One way to think about it is that the entry in the ith row and j^{th} column will come from the ith row of the left matrix and the jth column of the right matrix.

What about BA?

Section 5.1 Matrices and Linear Systems

In General $AB \neq BA$ for matricies!

Matrices of different sizes

```
octave:2> A=[1,2,3,4;5,6,7,8]
A =

   1   2   3   4
   5   6   7   8

octave:3> B=[1 2; 3 4; 5 6; 7 8]
B =

   1   2
   3   4
   5   6
   7   8

octave:4> A*B
ans =

    50    60
   114   140

octave:5> B*A
ans =

   11   14   17   20
   23   30   37   44
   35   46   57   68
   47   62   77   92
```

Matrices of the same size

```
octave:6> A=[1,2,3;4,5,6;7,8,9]
A =

   1   2   3
   4   5   6
   7   8   9

octave:7> B=[9,8,7;6,5,4;3,2,1]
B =

   9   8   7
   6   5   4
   3   2   1

octave:8> A*B
ans =

    30    24    18
    84    69    54
   138   114    90

octave:9> B*A
ans =

    90   114   138
    54    69    84
    18    24    30
```

Notice here they were the same, but this was **not** matrix multiplication. In Matlab or Octave * and .* (dot times) are very different. The .* operator multiplies component wise $a_{11}b_{11}$ $a_{12}b_{12}$ etc. In fact it is the Hadamard product we mentioned when first talking about vectors. If the matricies are of different sides you cannot do the Hadamard product.

```
octave:10> A.*B
ans =

    9   16   21
   24   25   24
   21   16    9

octave:11> B.*A
ans =

    9   16   21
   24   25   24
   21   16    9
```

Section 5.1 Matrices and Linear Systems

We stated in Theorem 3.11 that $A + 0_{nm} = A$ the zero matrix is often refered to as the additive identity for matrix addition, one might wonder then, is there a mltiplicative identity for matrix multiplication?

The identity matrix: Let $A \in R^{n\times m}$ then the identity matricies will be $I_m \in R^{m\times m}$, $I_n = R^{n\times n}$ and will have ones on the diagonal and zeros everywhere else. That is:

$$I = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \vdots \\ \vdots & 0 & \ddots & 0 \\ 0 & \cdots & 0 & 1 \end{bmatrix}$$ Further $AI_m = I_n A = A$

If $A \in R^{n\times n}$, is a square matrix $n = m$, then $I = \ I_n = I_m \in R^{n\times n}$ and we can say $AI = IA = A$

Example:

Multiply:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

.

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Section 5.1 Matrices and Linear Systems

Theorem Let s be a scalar and A, B, C be matrices of appropriate sizes. Then

$(a)\ A(BC) = (AB)C$
$(b)\ A(B + C) = AB + AC$
$(c)\ (A + B)C = AC + BC$
$(d)\ s(AB) = (sA)B = A(sB)$
$(e)\ AI = A$
$(f)\ IA = A$

Theorem Let A, B, C be nonzero matrices then,
(a) In general $AB \neq BA$
(b) $AB = 0$ **does not** imply that $A = 0$ or $B = 0$.
(c) $AC = BC$ **does not** imply that $A = B$ or $C = 0$

Example $\begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}\begin{bmatrix} -4 & 2 \\ 2 & -3 \end{bmatrix}$

Example Let $A = \begin{bmatrix} -3 & 3 \\ 11 & -3 \end{bmatrix}, B = \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$ Find AB *and* BC

Section 5.1 Matrices and Linear Systems

The Transpose of a Matrix

The transpose of a matrix is easy, columns become rows and rows become rows become columns

The first column becomes the first row, the second column becomes the second row, the third column becomes the third row and so on.

$$A=\begin{bmatrix}1&2&3\\4&5&6\\7&8&9\end{bmatrix} \xrightarrow{\text{Transpose...}} A^T=\begin{bmatrix}1&4&7\\2&5&8\\3&6&9\end{bmatrix}$$

Notice the diagonal stayed the same! Infact we are just moving a_{ij} *to* a_{ji} so on the diagonal $a_{ii}=a_{ii}$

Theorem 3.15: Let $A, B \in R^{n\times m}, C \in R^{m\times k}$ and s scalar. Then

(a) $(A+B)^T = A^T + B^T$
(b) $(sA)^T = sA^T$
(c) $(AC)^T = C^T A^T$

Short Proof of (a) good enough for quiz credit as well

Let $A+B=C$ with $A=[a_{ij}]$ $B=[b_{ij}]$ $C=[c_{ij}]=[a_{ij}+b_{ij}]$
Then Let $A^T=[a^T_{ij}]$, $B^T=[b^T_{ij}]$, $C^T=[c^T_{ij}]$, by the definition of transposition $a^T_{ij}=a_{ji}$, $b^T_{ij}=b_{ji}$, $C^T_{ij}=c_{ji}=a_{ji}+b_{ji}$.
Since $A^T+B^T=[a^T_{ij}+b^T_{ij}]=[a_{ji}+b_{ji}]=[c_{ji}]=[C^T_{ij}]$
and $[C^T_{ij}]=[a_{ij}+b_{ij}]^T=(A+B)^T \Rightarrow A^T+B^T=(A+B)^T$

Quiz problem: Prove (a)
We prove this by (1) calculating $A^T + B^T$ and (2) calculating $(A+B)^T$ and showing we get the same thing! Note tht we are fixing the values of the components in A and B and just moving them, so after the transposition the subscripts no longer correspond to the position in the matrix, rather the value itself. So if a_{32} was in row 3 col 2 in A and a^T_{23} in row 2 col 3 in A^T then $a^T_{23}=a_{32}$ which we put in the spot of a^T_{23}. I think the colors make things clear.

(1)

$$A=\begin{bmatrix}a_{11}&a_{12}&\cdots&a_{1n}\\a_{21}&a_{22}&\cdots&a_{2n}\\ \vdots&\vdots&&\vdots\\a_{m1}&a_{m2}&\cdots&a_{mn}\end{bmatrix},\ B=\begin{bmatrix}b_{11}&b_{12}&\cdots&b_{1n}\\b_{21}&b_{22}&\cdots&b_{2n}\\ \vdots&\vdots&&\vdots\\b_{m1}&b_{m2}&\cdots&b_{mn}\end{bmatrix} \xrightarrow{\text{add}} A+B=\begin{bmatrix}a_{11}+b_{11}&a_{12}+b_{12}&\cdots&a_{1n}+b_{1n}\\a_{21}+b_{21}&a_{22}+b_{22}&\cdots&a_{2n}+b_{2n}\\ \vdots&\vdots&&\vdots\\a_{m1}+b_{m1}&a_{m2}+b_{m2}&\cdots&a_{mn}+b_{mn}\end{bmatrix}$$

columns to rows (for A) and columns to rows (for B)

$$A^T=\begin{bmatrix}a_{11}&a_{21}&\cdots&a_{m1}\\a_{12}&a_{22}&\cdots&a_{m2}\\ \vdots&\vdots&&\vdots\\a_{1n}&a_{2n}&\cdots&a_{mn}\end{bmatrix},\ B^T=\begin{bmatrix}b_{11}&b_{21}&\cdots&b_{1n}\\b_{12}&b_{22}&\cdots&b_{2n}\\ \vdots&\vdots&&\vdots\\b_{1n}&b_{2n}&\cdots&b_{mn}\end{bmatrix} \xrightarrow{\text{add}} A^T+B^T=\begin{bmatrix}a_{11}+b_{11}&a_{21}+b_{21}&\cdots&a_{m1}+b_{m1}\\a_{12}+b_{12}&a_{22}+b_{22}&\cdots&a_{m2}+b_{m2}\\ \vdots&\vdots&&\vdots\\a_{1n}+b_{1n}&a_{2n}+b_{2n}&\cdots&a_{mn}+b_{mn}\end{bmatrix}$$

!! Same !

(2)

$$A+B=\begin{bmatrix}a_{11}+b_{11}&a_{12}+b_{12}&\cdots&a_{1n}+b_{1n}\\a_{21}+b_{21}&a_{22}+b_{22}&\cdots&a_{2n}+b_{2n}\\ \vdots&\vdots&&\vdots\\a_{m1}+b_{m1}&a_{m2}+b_{m2}&\cdots&a_{mn}+b_{mn}\end{bmatrix} \xrightarrow{\text{columns to rows}} (A+B)^T=\begin{bmatrix}a_{11}+b_{11}&a_{21}+b_{21}&\cdots&a_{m1}+b_{m1}\\a_{12}+b_{12}&a_{22}+b_{22}&\cdots&a_{m2}+b_{m2}\\ \vdots&\vdots&&\vdots\\a_{1n}+b_{1n}&a_{2n}+b_{2n}&\cdots&a_{mn}+b_{mn}\end{bmatrix}$$





Section 5.1 Matrices and Linear equations

The determinant of a matrix is a way to map all of the entries of a matrix to a single number. This single number can be used to determine whether or not a square matrix is invertible or to give us a sense of how much error we might see in a numerical calculation involving matrices.

We build up the definition of the determinant by first defining the easiest ones to calculate.

If $A = [a_1] \in R$ is a 1×1 matrix then the $\det(A) = |A| = a_1$

If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is a 2×2 matrix then $\det(A) = |A| = ad - bc$

If you recall the inverse of a 2×2 matrix was

$$A^{-1} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

and it was exactly when $(ad - bc) = 0$ that the matrix was not invertable. Infact:

An $n \times n$ matrix A is invertable if and only if $\det(A) \neq 0$

If $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ then the determinant is given by,

$$\det(A) = |A| = a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} - a_{13}a_{22}a_{31}$$

Formulas for determinants of larger matrices can be written down, but they get increasingly messy. On the next page we will learn a procedural way to calculate a determinant which relies on the definitions for smaller matrices here.

Is the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 0 & 0 & 0 \end{bmatrix}$ invertable?

Section 5.1 The Determinant Function

Let A be the $n \times n$ matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nm} \end{bmatrix}$$

Then the determinant of A is

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}$$

Cofactor Definition for Reference
Let A be an $n \times n$ matrix the cofactor C_{ij} of if the ij^{th} $entry$ a_{ij} is
$C_{ij} = (-1)^{i+j}\det(M_{ij})$
where M_{ij} is the submatrix formed by omitting the entries in the ith row and the jth column of A

Ex Let $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 6 \\ 7 & -1 & 1 \end{bmatrix}$
Find det(A)

In the calculation above it was nice that we had a 0 in the first row as it reduced the number of calculations.

Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 0 & 1 & 6 \\ 0 & -1 & 1 \end{bmatrix}$$

There are a lot of zeros in that first column! Using the definition above we would have expanded along the first row, however we can actually expand across any row or column.

Let A be an $n \times n$ matrix then

(a) $\det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}$ (Expansion across row i)
(b) $\det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}$ (Expansion across row j)

(a) *and* (b)*are called the cofactor expansions*

Section 5.1 Matrices and Linear Systems

Powers of a matrix

Just like we would say x multiplied by itself n times is written x^n we can do the same with a matrix

$$A^n = AA\cdots A \text{ where } A \text{ is multiplied by itself } n \text{ times}$$

Diagonal matrices are particularly nice for this operation.

$$A^n = \begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & 0 \\ 0 & 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix}\begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & 0 \\ 0 & 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix}\cdots\begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & 0 \\ 0 & 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix} = \begin{bmatrix} a_{11}^n & 0 & 0 & \cdots & 0 \\ 0 & a_{22}^n & 0 & \cdots & 0 \\ 0 & 0 & a_{33}^n & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn}^n \end{bmatrix}$$

Ex: Calculate A^3 if $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

Another important form of matrices are Upper Triangular and Lower Triangular matrices.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a_{22} & a_{23} & \cdots & a_{2n} \\ 0 & 0 & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix}$$

Upper Triangular

$$\begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ a_{21} & a_{22} & 0 & \cdots & 0 \\ a_{31} & a_{32} & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix}$$

Lower Triangular

Note that an Upper triangular matrix corresponds to echelon form and a diagonal matrix is almost in reduced row echelon form. We can also make the following statements.

(a) *Upper and Lower Triangular matrices are called triangular*
(b) *A Diagonal matrix is both Upper Triangular and Lower Triangular*
(c) *If A and B are both either Upper Triangular or Lower Triangular so is* $A + B$
(d) *Powers of triangular matricies are also triangular*

Section 5.1 Matrices and Linear Systems

We define the inverse matrix an $n \times n$ matrix A to be the matrix A^{-1} such that

$$A^{-1}A = I = AA^{-1}$$

Example: Let $A = \begin{bmatrix} 1 & 3 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} -5 & 3 \\ 2 & -1 \end{bmatrix}$ Show $B = A^{-1}$

Use the fact that $B = A^{-1}$ to solve the system: $x_1 + 3x_2 = 1$ and $2x_2 + 5x_2 = 2$

Section 5.1 Matrices and Linear Systems

Some nice formulas for nice matricies.

If a 2×2 matrix $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ in invertable then $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$

$$A^{-1}A = \frac{1}{ad-ac}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \frac{1}{ad-ac}\begin{bmatrix} ad-c & 0 \\ 0 & ad-bc \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

From this formula, we can tell A would not have an inverse if $ad - bc = 0$, we can't divide by zero!

If a diagonal matrix $A = \begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & 0 \\ 0 & 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix}$ in invertible then $A^{-1} = \begin{bmatrix} 1/a_{11} & 0 & 0 & \cdots & 0 \\ 0 & 1/a_{22} & 0 & \cdots & 0 \\ 0 & 0 & 1/a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1/a_{nn} \end{bmatrix}$

$$\begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & 0 \\ 0 & 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix}\begin{bmatrix} 1/a_{11} & 0 & 0 & \cdots & 0 \\ 0 & 1/a_{22} & 0 & \cdots & 0 \\ 0 & 0 & 1/a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1/a_{nn} \end{bmatrix}$$

$$= \begin{bmatrix} a_{11}/a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22}/a_{22} & 0 & \cdots & 0 \\ 0 & 0 & a_{33}/a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn}/a_{nn} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

Here we could see that if any of the $a'_{ii}s$ are zero the diagonal matrix would not have an inverse.